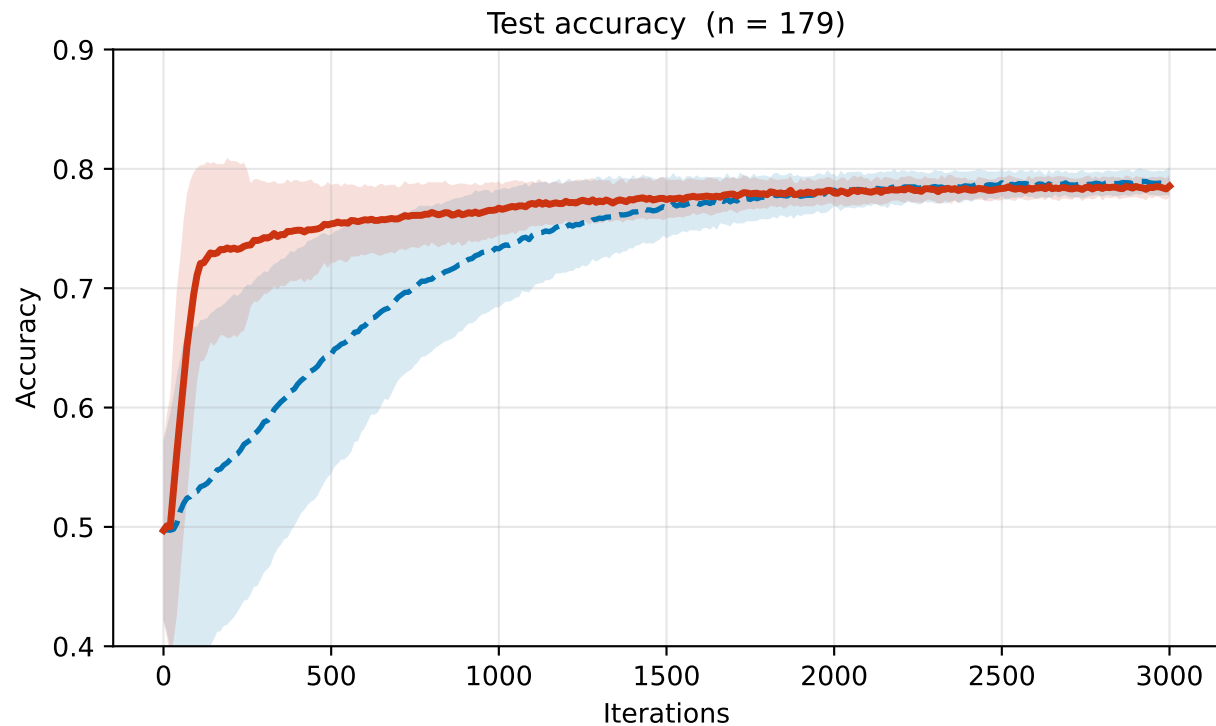
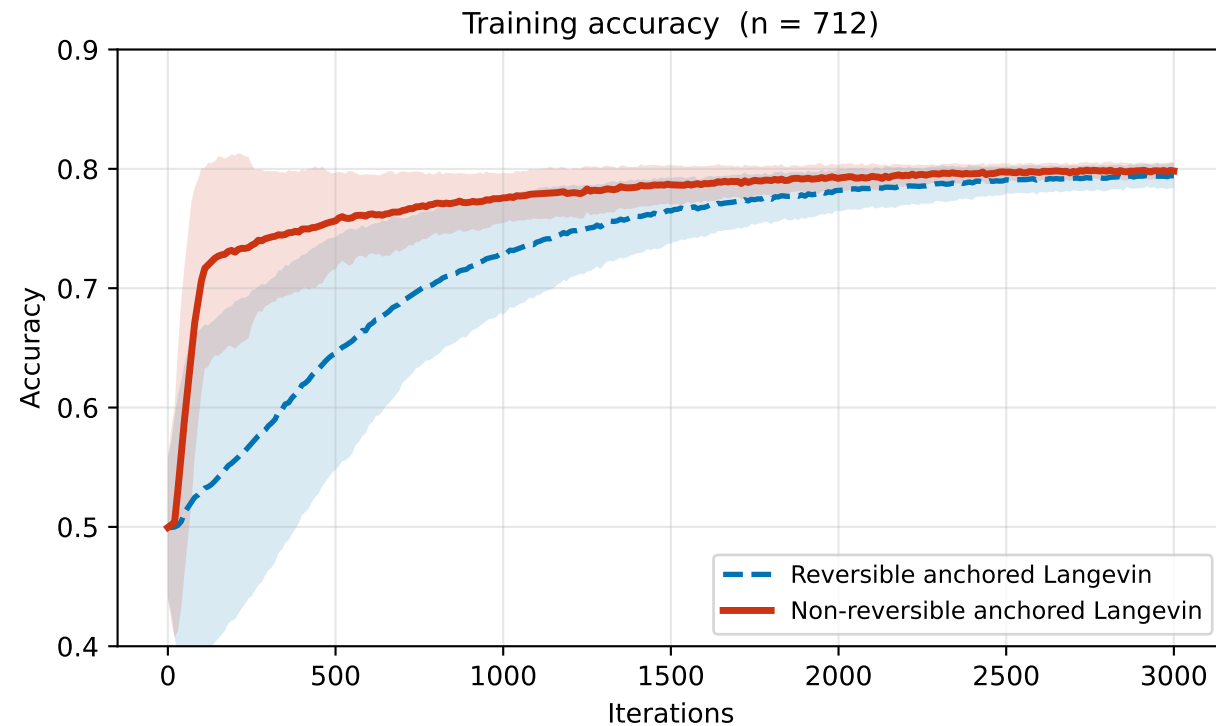


# Titanic — non-reversible vs reversible anchored Langevin (l1 constraint)



d = 9 (intercept + 8 columns: emb\_C, emb\_Q, sex, pclass, age, sibsp, log\_fare, parch); n\_train = 712, n\_test = 179;  $U = f + g$ ,  $g = 2 \cdot \sum |w_{jj}|$ , anchor delta = 0.02; EXACT gradient; eta = 7.0e-06; s = 8; R = 100. Preprocessing: scaling = 'design', order = 'design', signs aligned = True, block reparametrisation (v\_axis, v\_fast, v\_slow) = (1, 256, 1), (b, e) = (1.0, 0.5). Reference (scikit-learn logistic regression) test accuracy 0.782. Lines: across-replicate mean; bands: mean +/- 1 sd of single-iterate accuracy (repeat-run variability). Shared initialisation and noise per replicate. y-axis fixed to (0.4, 0.9), not windowed.